



# MEAM 5020 Final Project

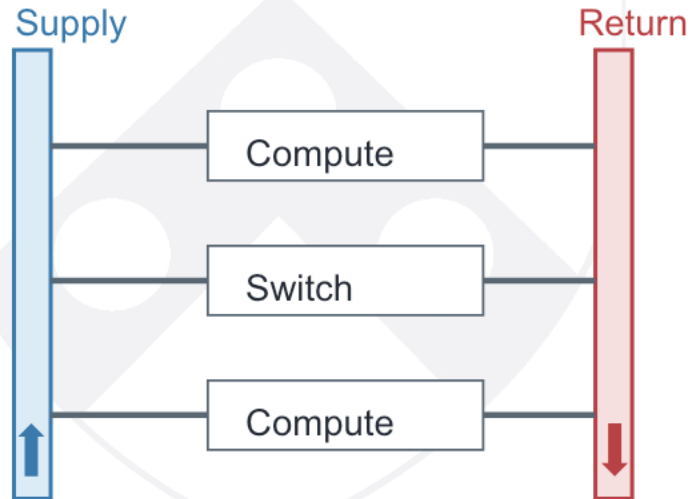
## Rack Manifold Optimization

---

Constantin de Lint, David Kim, Jason Simon, Sam Vasington

Assignment 4: Previous work, design changes, and calculation plan

# Coolant Distribution Across an AI Rack



**18 compute trays + 9 switch trays**  
**72 GPUs + 36 CPUs**

## Purpose

Deliver the cooling flow each tray needs while limiting pressure loss and pump power.

## How it works

A supply header feeds parallel tray branches. Cold plates absorb heat; a return header carries it to the CDU heat exchanger.

Different heat loads and flow resistances create different cooling needs.

[4] [7] Sources and calculation details in speaker notes.

# Three Approaches From Previous Work

| Previous approach                                              | Evidence                                                        | Use in this project                                    |
|----------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------|
| [1] Patterned branch orifices<br>Sogunuru et al., 2020         | Electronics cooling with unequal flow targets                   | Selected baseline: tune branch resistance to heat load |
| [2] Tapered header<br>Hassan et al., 2014                      | At $Re = 150,000$ , reported $\Phi$ falls from 0.0345 to 0.0140 | Compare header taper against added restrictions        |
| [3] Header shape and inlet<br>Anbumeenakshi & Thansekhar, 2016 | 25 water microchannels; best shape depends on flow rate         | Test operating range before selecting geometry         |

[1] [2] [3] Sources and calculation details in speaker notes.

# Selected Baseline: Patterned Orifice Sizing

| Published setup [1] | Value                                      |
|---------------------|--------------------------------------------|
| Configuration       | Z-type, water, 1 L/min total               |
| Header / branch ID  | 12 mm / 6.5 mm                             |
| Channels / length   | 10 channels / 700 mm                       |
| Pattern 1 example   | 2.1–2.4 mm bores in branches 1, 3, 5, 7, 9 |

## Why this baseline?

It directly links unequal electronics heat loads to individually sized restrictions.

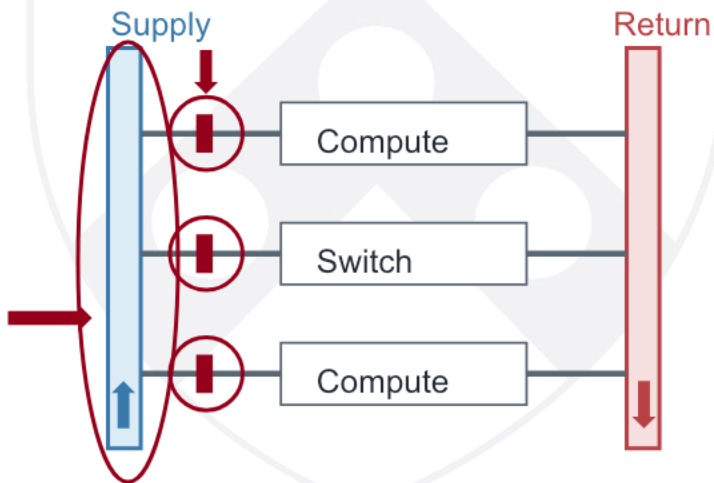
## Our extension

Couple 27 tray branches to temperature-dependent properties, component losses, and CDU limits.

Rack geometry is assumed; reproducing the literature benchmark is a future validation step.

[1] [7] Sources and calculation details in speaker notes.

# Physical Changes to the Manifold



Schematic, not to scale

## Replaceable orifice inserts

Set compute and switch bores separately; allow a different bore at each tray.

## Header diameter and taper

Adjust the main passage to manage pressure variation along the rack.

## Branch and QD passage sizes

Vary flow area; verify with measured loss curves.

[1] [7] Sources and calculation details in speaker notes.

# Goal and Design

| Project item           | Definition                                                                                                            |
|------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Problem / project type | Unequal cooling allocation / component design                                                                         |
| Primary goal           | Match each tray flow to its heat load                                                                                 |
| Proposed target        | RMS flow-target error $\leq 5\%$ at the design point                                                                  |
| Design variables       | Header ID/taper; compute, switch, and per-tray bores                                                                  |
| Secondary objectives   | Minimize outlet temperature, pump power, and header volume                                                            |
| Study constraints      | Outlet $\leq 65\text{ }^{\circ}\text{C}$ ; header velocity $\leq 3\text{ m/s}$ ; $0 < \text{bore} < \text{branch ID}$ |
| Decision rule          | Check fixed hardware and pump capability off-design                                                                   |

[7] Sources and calculation details in speaker notes.

# Rack Calculation Inputs and Assumptions

| Input                        | Study value                                          | Evidence status                |
|------------------------------|------------------------------------------------------|--------------------------------|
| Rack topology                | 18 compute + 9 switch trays                          | Published topology [4]         |
| Liquid heat scenario         | $18 \times 5.80 + 9 \times 1.24 = 115.56 \text{ kW}$ | Model load assumptions         |
| Rack coolant / supply / flow | Legacy PG25 / 40 °C / 120 L/min                      | Chosen operating point         |
| Headers / branch IDs         | 38 mm / compute 8 mm, switch 6 mm                    | Assumed internal diameters     |
| QD and cold-plate losses     | Separate reduced resistance coefficients             | Estimated, not measured curves |
| Facility water               | 36 °C, 150 L/min, one rack                           | Scenario, not a facility limit |

[4] [7] Sources and calculation details in speaker notes.

# Mass Balance and Heat-Proportional Flow

## 1. Conserve coolant at every junction

$$\Sigma \dot{m}_{in} = \Sigma \dot{m}_{out} \quad \text{and} \quad \Sigma \dot{m}_i = \dot{m}_{rack}$$

## 2. Allocate flow according to heat

$$\dot{m}_i^* = \dot{m}_{rack} \times \dot{Q}_i / \Sigma \dot{Q}_i$$

At 120 L/min: compute target = 6.02 L/min per tray; switch = 1.29 L/min.

## 3. Solve each tray energy balance

$$\dot{Q}_i = \dot{m}_i [h(T_{out,i}) - h(T_{in})] \quad h(T) = \int c_p(T) dT$$

$$\text{Flow error (\%)} = 100 \sqrt{\{(1/n) \Sigma [(\dot{m}_i / \dot{m}_i^*) - 1]^2\}}$$

[7] Sources and calculation details in speaker notes.



# Pressure Loss and Orifice Sizing

## Pipe and fitting losses

$$\Delta p = (f L/D + \Sigma K) \rho v^2/2$$

$$Re = \rho v D / \mu$$

Use  $64/Re$  in laminar flow; a Haaland relation in turbulent flow, with a transition blend.

## Network closure

Each complete supply–tray–return path has the same end-to-end pressure difference. Include elevation head.

## Permanent orifice loss

$$\Delta p_{\text{orifice}} = K_h \dot{m}^2$$

$$\beta = d/D \quad A_o = \pi d^2/4$$

$$K_h = [\sqrt{(1 - \beta^4(1 - C_d^2))} - C_d \beta^2]^2 / (2\rho C_d^2 A_o^2)$$

For required  $K_h$ , invert this relation to obtain bore  $d$ . Smaller bores increase resistance.

$0 < d < D$ ;  $C_d = 0.62$  is assumed.

[7] Sources and calculation details in speaker notes.

# Pump and Facility Heat-Exchanger Balance

## Pump operating point and electrical power

$$\Delta p_{\text{pump}} = \Delta p_0 N^2 - a \dot{V}^2 = \Delta p_{\text{system}} \quad P_e = \Delta p \dot{V} / \eta$$

Fixed-flow mode evaluates the head required. Pump mode solves the curve intersection.

## Facility water removes the rack heat

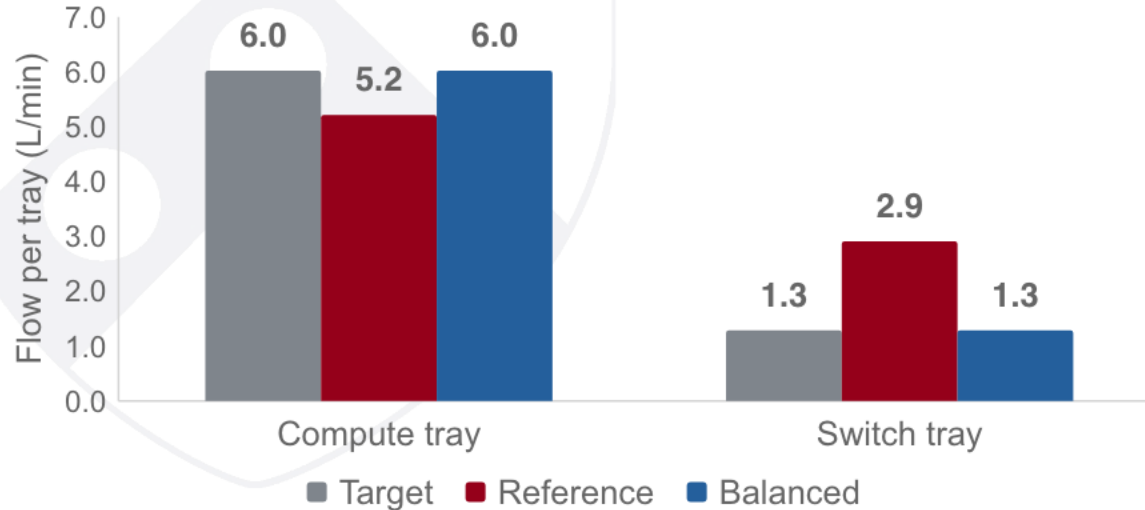
$$\dot{Q}_{HX} = \sum \dot{Q}_i = \dot{m}_{FWS} [h(T_{FWS,out}) - h(T_{FWS,in})]$$

## Heat-exchanger conductance required

$$UA_{\text{required}} = \dot{Q}_{HX} / \Delta T_{lm} \quad \Delta T_{lm} = (\Delta T_1 - \Delta T_2) / \ln(\Delta T_1 / \Delta T_2)$$

[7] Sources and calculation details in speaker notes.

# Flow Redistribution Matches Cooling Demand



## Same total flow

Restricting over-supplied switch branches redirects coolant to compute trays.

Equal flow is not the objective: the modeled compute heat load is 4.68× larger.

[7] Sources and calculation details in speaker notes.

# Better Distribution Requires More Pumping

| Metric                         | Reference | Balanced   |
|--------------------------------|-----------|------------|
| RMS flow-target error          | 73.49%    | 0.44%      |
| Maximum tray outlet            | 55.97 °C  | 53.80 °C   |
| Tray outlet temperature spread | 9.90 K    | 0.21 K     |
| Required system pressure       | 79.83 kPa | 105.09 kPa |
| Modeled pump electrical power  | 266.1 W   | 350.3 W    |
| Mixed rack return temperature  | 53.73 °C  | 53.73 °C   |

**Maximum outlet falls 2.17 K; pumping rises 84.2 W (+31.6%).**

Heat and total flow are unchanged, so the mixed return stays the same.

[7] Sources and calculation details in speaker notes.

# Existing Code and Data to Reuse

| Resource                     | Role                                                        | Status               |
|------------------------------|-------------------------------------------------------------|----------------------|
| SciPy least_squares [6]      | Solve nonlinear hydraulic residuals with bounds             | Already used         |
| Project Python model [7]     | Network, coolant properties, sweeps, constraints, Streamlit | Already implemented  |
| CoolProp INCOMP database [5] | Independent water/glycol property check                     | Proposed cross-check |

## Calculation workflow

Solve flows and temperatures, update properties, and iterate.  
Then check constraints and compare designs.

[5] [6] [7] Sources and calculation details in speaker notes.

# Checks Needed Before Choosing Hardware

---

## **Numerical checks passed**

Both example cases converge and satisfy mass, energy, and pressure closure.

## **Validate the inputs that drive the answer**

Measure tray/QD pressure–flow curves and orifice discharge behavior. Obtain chip limits, thermal resistance, and facility/CDU performance maps.

## **Test fixed hardware across operating conditions**

Sweep flow, workload split, coolant, and inlet temperature. Include bore tolerances, partial blockage, and reduced pump capability.

The 1D model omits 3D tee momentum recovery. Verify header-shape benefits with CFD.

[7] Sources and calculation details in speaker notes.

# References and Calculation Record

---

- [1] Sogunuru et al. (2020). Patterned flow distribution through manifolds in electronics cooling. IJMPERD 10(2), 975–986. Selected calculation baseline.
- [2] Hassan et al. (2014). Modeling the Uniformity of Manifold with Various Configurations. Journal of Fluids. doi:10.1155/2014/325259.
- [3] Anbumeenakshi & Thansekhar (2016). Header shape and inlet configuration in microchannels. Experimental Thermal and Fluid Science 75, 156–161. doi:10.1016/j.expthermflusci.2016.02.004.
- [4] NVIDIA DGX GB Rack Scale Systems User Guide: Hardware.
- [5] CoolProp: Incompressible Fluids database and documentation.
- [6] SciPy: `scipy.optimize.least_squares` documentation.
- [7] NVL72-Manifold-Optimization repository. Matched reference and balanced runs recalculated on 15 September 2026. Full source links and details are in slide notes.